

Data Scientist Take Home Questions

Introduction

In this case study, you will be exposed to problems that you would solve in any typical work day as a data scientist. For the purposes of this role, we will be assessing you on:

- Coding skills in Python and ability to deal with relatively large datasets
- Exploratory Data Analysis and Feature Engineering
- Ability to apply the right algorithm(s) to the problem sets
- Ability to improve model performance

Deliverables

- Your Python code
- A presentation (5-7 slides), where you can provide a brief explanation of the different techniques that you have used and your thought process.

Case Study:

A Company offers users to purchase digital contents through a variety of payment options, called payment channels. The payment channels range from direct carrier billing, where the user pays with the mobile phone balance, e-wallets, credit and debit cards, etc. Each of the payment channels are susceptible to fraudulent activities, such as stolen credit card, phishing the one time password for credit card payments, unauthorized usage of e-wallets by family members, etc. Fraudulent activities can lead to loss of revenue and loss of reputation for the Company, therefore, it is essential to identify potential fraudulent transactions. You are provided with historical transaction level data with flags showing which of the transactions are reported by the user of the payment service as fraudulent. Your task is to build a model that predicts if a transaction will be flagged by a user. The data and its description are attached in this zipped file, with the file named respectively 'sample_dataset_data_scientist.xlsx' and 'description_of_the_dataset'.